

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

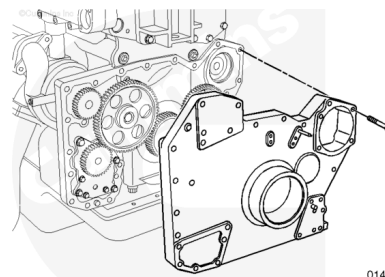
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries and air starter to prevent accidental starting.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>).
- Remove the fan and fan spacer. Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Remove the fan belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Remove the fan belt tensioner. Refer to Procedure 008-087 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html>).
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>).
- Remove the alternator drive gear and shaft. Refer to Procedure 009-009 in Section 9 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>).
- Drain the lubricating oil from the oil pan. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the front engine support bracket. Refer to Procedure 016-002 in Section 16 (</qs3/pubsys2/xml/en/procedures/89/89-016-002.html>).

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : One capscrew must be removed from the backside of the cover, located in the bottom of the fuel pump support plate.

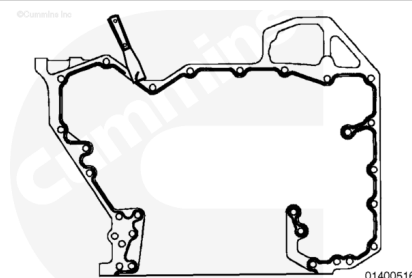
Remove the front gear cover capscrews.

Remove the front gear cover.

Clean

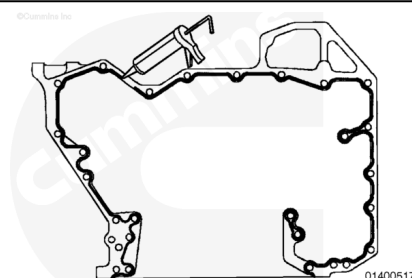
Clean the gasket sealant from the front gear cover and the block.

Inspect and clean off the gasket sealant from tapped blind holes, if present.



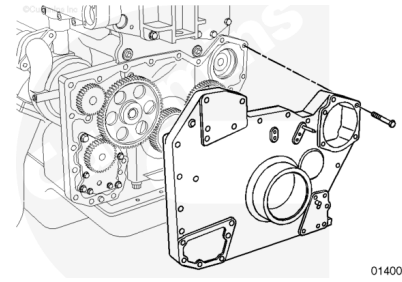
Install

Apply a bead of gasket sealant 2 to 3 mm [0.079 to 0.118 in] wide to the front cover.



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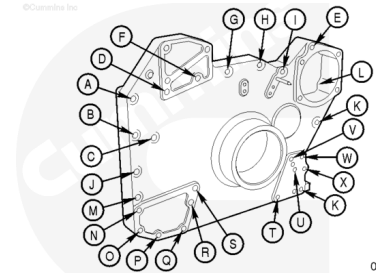


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Install the front gear cover.

Install the front cover capscrews.

Several different lengths of capscrews are used to mount the front cover. See the illustration to find the appropriate capscrew length for each location. Some of the capscrews mount other components in addition to the front cover.



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Capscrew Location	Length (Industrial)	Length (Power-Generation)
A, B, C, D, E	115 mm [4.5 in]	115 mm [4.5 in]
F, G, H, I, J, K	75 mm [2.95 in]	75 mm [2.95 in]
L, M	65 mm [2.6 in]	65 mm [2.6 in]
N	Location not used	40 mm [1.6 in]
O	75 mm [2.95 in]	100 mm [3.9 in]
P	75 mm [2.95 in]	100 mm [3.9 in]
Q	75 mm [2.95 in]	100 mm [3.9 in]
R	Location not used	100 mm [3.9 in]
S	75 mm [2.95 in]	100 mm [3.9 in]
T	Location not used	40 mm [1.6 in]
U	Location not used	100 mm [3.9 in]
V	75 mm [2.95 in]	100 mm [3.9 in]
W	75 mm [2.95 in]	100 mm [3.9 in]
X	75 mm [2.95 in]	100 mm [3.9 in]
Y	75 mm [2.95 in]	100 mm [3.9 in]

Tighten the capscrews.

Torque Value: 113 n•m [83 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the alternator drive gear and shaft. Refer to Procedure 009-009 in Section 9.
(/qs3/pubsys2/xml/en/procedures/89/89-009-009.html)
- Install the front crankshaft seal. Refer to Procedure 001-023 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-023.html).
- Install the vibration damper. Refer to Procedure 001-052 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-052.html).
- Install the fan belt tensioner assembly. Refer to Procedure 008-087 in Section 8
(/qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html).
- Install the alternator drive idler gear. Refer to Procedure 001-097 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-097.html).
- Install the alternator. Refer to Procedure 013-001 in Section 13
(/qs3/pubsys2/xml/en/procedures/89/89-013-001.html).
- Install the fan belt. Refer to Procedure 008-002 in Section 8
(/qs3/pubsys2/xml/en/procedures/89/89-008-002.html).
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 in Section 8
(/qs3/pubsys2/xml/en/procedures/89/89-008-040.html).
- Install the front engine support bracket. Refer to Procedure 016-002 in Section 16
(/qs3/pubsys2/xml/en/procedures/89/89-016-002.html).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html).
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8
(/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html).
- Connect the batteries.
- Operate the engine to normal operating temperature. Check for leaks.